

Detected Ice → Reachable Mission Targets

The ice-detection stage does not directly choose the landing point. It hands over high-evidence coordinates that become mission targets.

Ice Evidence Map

Output from multi-sensor ice detection



High-evidence coordinates / clusters

Handoff Logic

Ice coordinates

Latitude-longitude cells with strong evidence



Target clusters

Grouped targets after removing weak/noisy points



Nearby landing search

Safe zones around the target clusters

Theme: detect ice first, then land close enough to reach it.

SCORING MODEL

Best Site = Safe Landing + Reachable Ice

Shortlist landing zones only after safety filtering. Then rank them by how much high-evidence ice the rover can realistically reach.

Final Site Score

+ Landing Safety

+ Illumination

+ Communication

+ Thermal Survival

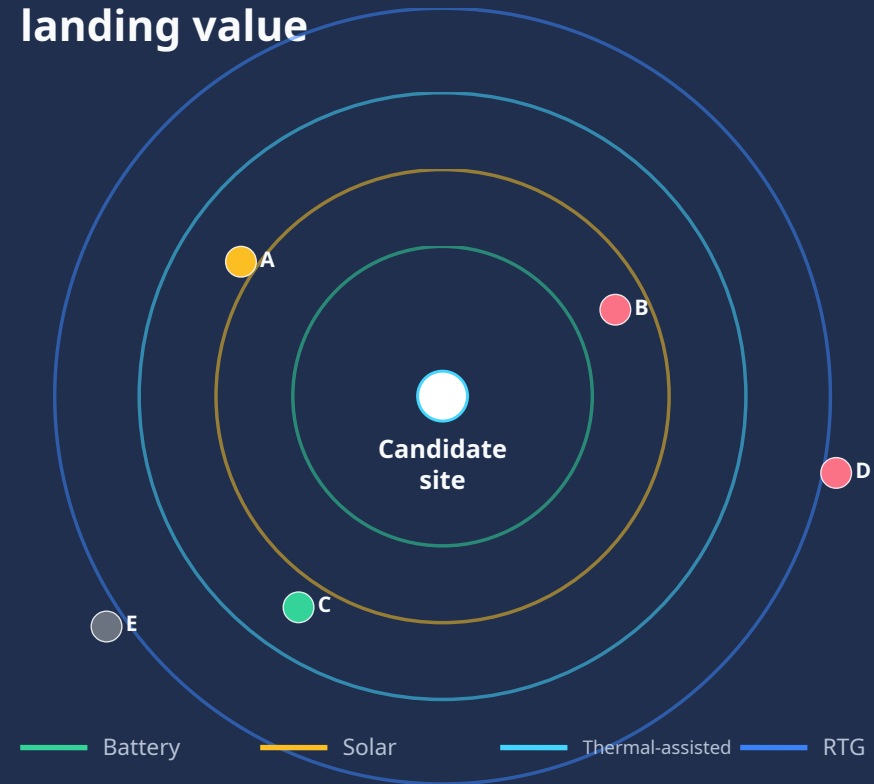
+ Reachable Ice Opportunity

– Risk Penalty

Reachable Ice Opportunity

$\Sigma (\text{Ice score} \times \text{Confidence} \times \text{Access} \times \text{Distance})$

Rover-class radius turns ice evidence into landing value



Selected site = safe touchdown zone with maximum realistic access to high-evidence ice.